

Equal Temperament and Diophantine Confinement: Why the Major Third Has Three Strands

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Abstract

We explain a striking visual pattern in the approximation of the just major third (frequency ratio $5/4$) by N -tone equal temperament: when the error $|2^{m/N} - 5/4|$ is plotted against N , the data separates into three distinct strands. The explanation is purely Diophantine: $\log_2(5/4) \approx 0.32193$ is well-approximated by $1/3$, so the residue $N \bmod 3$ controls which strand a given temperament occupies. We show that this is the one-dimensional analogue of the Diophantine Confinement phenomenon governing the Collatz map, where $\log_2 3$ plays the role of $\log_2 5$ and the torus sieve replaces the number line.

1 The problem: fitting a just third into equal temperament

In N -tone equal temperament (hereafter N -TET), the octave is divided into N equal logarithmic steps. The m -th scale degree produces the frequency ratio $2^{m/N}$. A *just major third* is the ratio $5/4 = 1.25$. The question is:

For each N , which integer m minimises $|2^{m/N} - 5/4|$, and how small can that error be?

Define

$$\alpha = \log_2(5/4) = \log_2 5 - 2 \approx 0.32192809489 \dots \quad (1.1)$$

Then the optimal m for a given N is $m^*(N) = \text{round}(N\alpha)$, and the error in the *exponent* is

$$\varepsilon(N) = \min_{m \in \mathbb{Z}} |N\alpha - m| = \|N\alpha\|, \quad (1.2)$$

where $\|\cdot\|$ denotes the distance to the nearest integer. The error in the *frequency ratio*, which is what the graph displays, is

$$E(N) = |2^{m^*(N)/N} - 5/4| = \frac{5}{4} |2^{\pm\varepsilon(N)/N} - 1| \approx \frac{5 \ln 2}{4} \frac{\varepsilon(N)}{N} \quad (1.3)$$

for small $\varepsilon(N)/N$. Both measures tell the same structural story.

2 Why three strands?

Observation 2.1 (Diophantine bottleneck at $1/3$). The continued fraction expansion of α begins

$$\alpha = [0; 3, 9, 2, 1, 1, 3, 1, 18, \dots],$$

with convergents $0, 1/3, 9/28, 19/59, \dots$. The very first convergent $p_1/q_1 = 1/3$ is already an excellent approximation:

$$\alpha - \frac{1}{3} = -0.01140\dots$$

This means that $\alpha \approx 1/3$ with relative error about 3.5%.

Because $\alpha \approx 1/3$, the nearest-integer problem for $N\alpha$ behaves very differently in the three residue classes modulo 3.

The $N \equiv 0$ strand (locked at $1/3$). When $N = 3k$, the nearest integer to $N\alpha \approx k - k\delta$ is $m = k$ (for moderate k), giving $m/N = 1/3$ exactly. The error is therefore *locked* at $E = |2^{1/3} - 5/4| \approx 0.00992$, forming a flat horizontal band at the top of the graph. This is why $N = 24$ gives the same error as $N = 12$: both use $m/N = 1/3$.

The $N \equiv 1$ strand (contains $q_2 = 28$). For $N = 3k + 1$, the product $N\alpha \approx k + \alpha - k\delta$ is close to an integer plus $\alpha \approx 0.322$. The rounding error $\|N\alpha\|$ is therefore small—and at the convergent denominator $N = 28$ it nearly vanishes ($9/28 \approx 0.32143$). This strand starts near zero and slowly rises as N grows.

The $N \equiv 2$ strand (contains $q_3 = 59$). For $N = 3k + 2$, the product $N\alpha \approx k + 2\alpha - k\delta$ is close to an integer plus $2\alpha \approx 0.644$. The rounding error $\|N\alpha\| \approx 1 - 2\alpha \approx 0.356$ is initially large, but it decreases as the drift $N\delta$ accumulates. At the next convergent $N = 59$, it hits a spectacular near-zero ($19/59 \approx 0.32203$).

Residue	m/N	Error near $N=24$	Trend	Key N
0 (mod 3)	$1/3$	0.0099 (locked)	flat, falls	84, 87
1 (mod 3)	$\approx \alpha$	0.0004 at 28	low, rises	28, 31
2 (mod 3)	$< \alpha$	0.0123 at 26	high, falls	56, 59

This creates three strands of dots that drift and eventually cross each other as the accumulated phase $N\delta$ grows.

Proposition 2.2 (Strand separation). *Let $\delta = 1/3 - \alpha \approx 0.01140$.*

1. *If $N = 3k$, then $N\alpha = k - 3k\delta$, so $m^* = k$ and $\varepsilon(N) = 3k\delta$ (growing linearly in k , but $E(N) = |2^{1/3} - 5/4|$ is constant because $m^*/N = 1/3$ regardless of k). This lock persists until $3k\delta > 1/2$, i.e. $k > 1/(6\delta) \approx 14.6$, so around $N = 45$ the strand “unlocks” and m^* shifts.*
2. *If $N = 3k+1$, then $\varepsilon(N) = |(3k+1)\alpha - m^*| \approx |k\alpha' + \alpha - m^*|$ where $\alpha' = 3\alpha - 1 \approx -0.0342$. The exponent error oscillates with period $1/|\alpha'| \approx 29$, hitting near-zero at multiples of the convergent denominator $q_2 = 28$.*
3. *If $N = 3k+2$, then $\varepsilon(N)$ starts large (≈ 0.356 at $k = 8$) and decreases as k grows, reaching near-zero at the convergent $q_3 = 59$.*

The strands are not static: because $\alpha \neq 1/3$, the accumulated phase $N\delta$ drifts, and eventually a strand “crosses” another. The $N \equiv 0$ strand descends from its locked plateau; the $N \equiv 1$ strand rises from its initial low; the $N \equiv 2$ strand falls from its initial high. A full cycle of strand reordering occurs roughly every $1/\delta \approx 88$ values of N —visible in the graph as the strands weave through each other.

3 The $N = 24$ stagnation

A natural question: if 12-TET approximates the major third with $m = 4$ (so $2^{4/12} = 2^{1/3}$), does doubling the resolution to 24-TET help?

Observation 3.1 ($N = 24$ stagnation). For $N = 24$, the optimal m is $m^*(24) = \text{round}(24\alpha) = \text{round}(7.7263) = 8$. But $8/24 = 1/3$, so

$$2^{8/24} = 2^{1/3} = 2^{4/12}.$$

The error is identical. Subdivision by a factor of 2 refines the lattice, but the new lattice points all miss α by the same amount because $2 \cdot 4 = 8$ simply doubles the old approximant without improving it.

More generally, if $N' = cN$ for a positive integer c , then $m^*(N') = cm^*(N)$ whenever $cm^*(N)$ remains the nearest integer to $N'\alpha$. This happens precisely when c does not push $N'\alpha$ past a half-integer—i.e., when the refinement does not “discover” a new lattice point between the old ones. Improvement requires reaching the denominator of the *next* convergent: $q_2 = 28$ (where $9/28$ gives a much better approximation than $1/3$).

4 Connection to Diophantine Confinement in the Collatz map

The Collatz map $T(n) = n/2$ (even), $T(n) = (3n + 1)/2$ (odd) generates trajectories whose growth is governed by the interplay of multiplication by 3 and division by 2. After t steps with ν_3 odd steps and ν_2 total halvings, the iterate satisfies

$$n_t \approx n_0 \cdot \frac{3^{\nu_3}}{2^{\nu_2}}.$$

The **deficit** at time t is

$$\Delta(t) = \nu_2(t) \log_2 2 - \nu_3(t) \log_2 3 = \nu_2(t) - \nu_3(t) \log_2 3, \quad (4.1)$$

and convergence to 1 requires $\Delta(t) \rightarrow +\infty$.

4.1 The structural parallel

The table below makes the analogy precise:

Feature	Musical thirds	Collatz dynamics
Irrational constant	$\alpha = \log_2(5/4)$	$\lambda = \log_2 3$
Lattice parameter	N (divisions of octave)	t (trajectory time)
Integer approximant	m (scale degree)	ν_3 (odd-step count)
Error / deficit	$\ N\alpha\ $	$\nu_2 - \nu_3\lambda$
Best rational approx	convergents of α	convergents of λ
Bottleneck fraction	$1/3$	$1/2$ (since $\log_2 3 \approx 1.585$)
Strand count	3 (from $1/3$)	—

In both settings, the core mechanism is identical:

1. An irrational number α (or λ) must be approximated by rationals m/N (or ν_3/ν_2).
2. The quality of approximation is governed by the continued fraction expansion, and the first convergent creates a *Diophantine bottleneck* that organizes the error into discrete strands.
3. Between convergent denominators, the error *stagnates* (cf. $N = 24$ vs. $N = 12$).
4. At convergent denominators, the error drops sharply (cf. $N = 28$ and $N = 59$ in the graph).

4.2 Cell error and the torus sieve

In the Collatz setting, the trajectory $(\nu_2(t), \nu_3(t))$ lives on the integer lattice $\mathbb{Z}^2$, and its distance from the “equilibrium line” $\nu_2 = \lambda \nu_3$ is the deficit (4.1). The torus sieve projects this lattice onto $(\mathbb{Z}/k\mathbb{Z})^2$ for successive moduli $k = 3, 9, 27, \dots$, and the **cell error** at resolution k is

$$e(a, b; k) = a - b\lambda \pmod{k}.$$

This is exactly the quantity $\|N\alpha\|$ from the musical problem, with k playing the role of N and λ replacing α .

The “three strands” of the musical graph correspond to the three cells of the $k = 3$ torus sieve: cells $(a, b) \bmod 3$ with $b \equiv 0, 1, 2$. The “dangerous cells” where the error is small are the musical *local minima* where N -TET nearly nails the just third.

4.3 Baker’s theorem as the moat

Baker’s theorem on linear forms in logarithms gives

$$|p \log 2 - q \log 3| > C(p + q)^{-\mu}$$

for an effective exponent μ . In the musical analogy, this is the statement that no N -TET can approximate $5/4$ *too* well:

$$\|N\alpha\| > c N^{-\mu'}$$

for the irrationality measure μ' of $\alpha = \log_2(5/4)$. The white space between strands in the graph—the “safe sea” where no dots appear—is the visual manifestation of this Diophantine lower bound. In the Collatz torus sieve, the analogous gap is the **Baker moat**: the minimum cell error below which no trajectory can fall, which forces every trajectory to eventually accumulate enough deficit to descend.

4.4 Strand crossing and kick vectors

As N increases, the three strands slowly drift because $\alpha \neq 1/3$. When the accumulated drift $N\delta$ exceeds $1/2$, the strand ordering permutes: what was the “low error” strand becomes the “high error” strand. In the Collatz setting, the analogous phenomenon is the **kick vector**: as the trajectory time t grows, the cell occupied at resolution k shifts, and the trajectory is “kicked” from a low-error cell to a high-error cell (or vice versa). The Baker bound ensures that after each kick, the trajectory spends enough time in high-error cells to accumulate positive drift before the next dangerous passage.

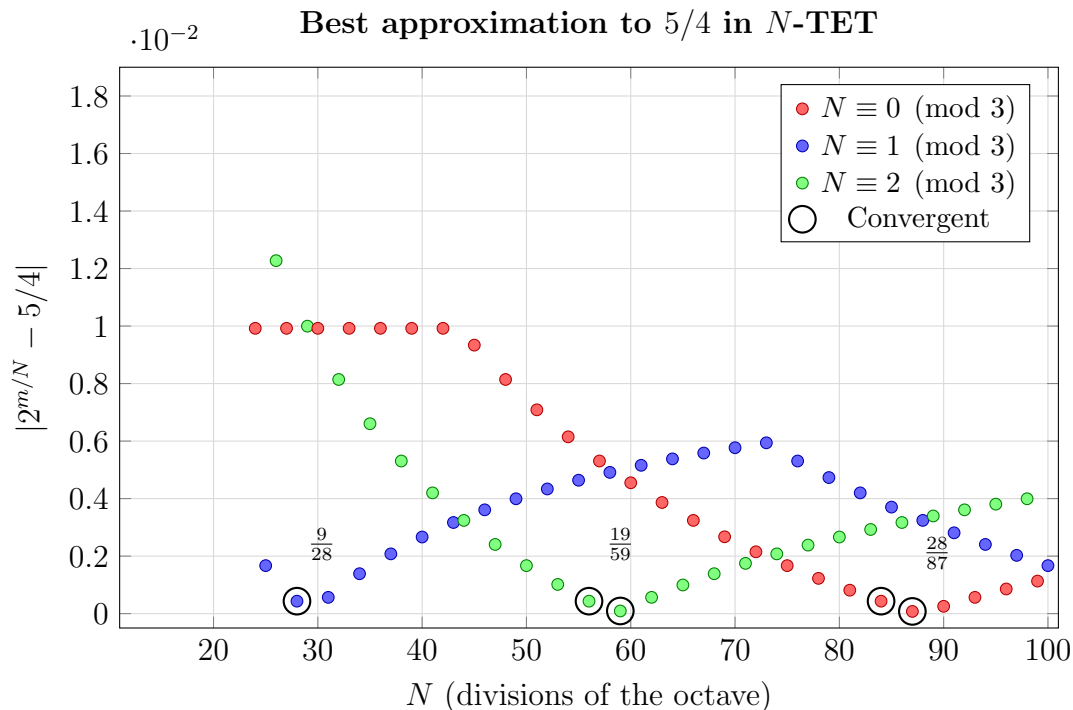


Figure 1: Error in best approximation to the just major third ($5/4$) in N -TET, colored by $N \bmod 3$. Circled points are convergent denominators of $\log_2(5/4)$. The three-strand structure arises because $\log_2(5/4) \approx 1/3$.

5 The graph

6 Summary

The three-strand pattern in the equal temperament graph is not a coincidence of music theory—it is a universal consequence of Diophantine approximation near a low-order rational. The same mechanism, with $\log_2 3$ replacing $\log_2(5/4)$, governs the cell-error structure of the Collatz torus sieve. In both cases:

- The continued fraction of the irrational constant determines where the “record” approximations fall (convergent denominators).
- The first convergent’s denominator determines how many strands ap-

pear (here 3, from $1/3$).

- Lattice refinements that are multiples of existing denominators produce no improvement (the $N = 24$ stagnation).
- Baker-type lower bounds on linear forms in logarithms guarantee that the error can never vanish, creating the “moat” between strands.

In the words of equal temperament: *the twelve-tone scale cannot perfectly tune the major third, because $\log_2 5$ is irrational and its continued fraction traps every multiple of 12 in the same strand.* In the words of the Collatz conjecture: *the trajectory cannot stagnate, because $\log_2 3$ is irrational and Baker’s theorem forces the deficit to grow.*

The graph is, in essence, a one-dimensional cross-section of the two-dimensional torus sieve.

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